

**Keagan Ryder**

## **5-1 Journal: Computer Science Trends and Artifact Update**

**December 7, 2025**

### **Part One**

Computer science continues to evolve rapidly, and two of the most influential emerging trends today are artificial intelligence automation using large language models (LLMs) and the increase in real-time cybersecurity driven by machine-learning threat detection. Both trends reflect where the field is heading and already influence educational expectations, workforce demands, and the nature of digital systems.

The significance of AI automation, particularly through advanced LLMs, lies in its ability to fundamentally change how software is developed, tested, and maintained. Instead of writing every component from scratch or executing long debugging cycles manually, developers increasingly rely on AI assistance to accelerate problem solving, automate documentation, generate code templates, and identify logical or security flaws. This trend will shift the field of computer science from being code-centric to being design-centric. The new emphasis will be understanding architectures, constraints, human-AI interaction, and ethical considerations rather than just syntax and implementation details. For consumers and workers, this will lead to software that is developed faster, updated more frequently, and increasingly customized. Citizens will interact with systems that anticipate needs, personalize experiences, and integrate natural-language interfaces into almost every device.

For my own career aspirations, this trend aligns directly with my background in software engineering, algorithms, and data-driven decision making. I am particularly interested in using AI to solve environmental and industrial challenges, such as automating monitoring systems or optimizing large-scale processes. Understanding the foundations of AI and how these tools integrate with traditional systems gives me a competitive advantage when stepping into roles that blend programming, intelligence systems, and real-world problem solving.

The second trend, real-time cybersecurity powered by machine learning, is significant because threats now evolve faster than traditional security systems can respond. Instead of relying solely on static rules, modern systems use anomaly detection, reinforcement models, and adaptive filtering to identify attacks instantly. This trend changes the field of computer science by pushing security to the forefront of all design considerations. Rather than treating cybersecurity as a final step, developers are now expected to build systems from the ground up with a security mindset.

Consumers will interact with devices that are safer, more predictive, and less vulnerable to widespread breaches. For workers and citizens, this means fewer disruptions, more secure services, and increased trust in digital infrastructure. As someone drawn to cybersecurity, industrial systems, and systems analysis, this trend aligns closely with my long-term interests. My computer science capstone work has strengthened my understanding of secure data structures, safe system design, and the importance of validating user input and database interactions. These foundations will carry into job roles that involve securing automation equipment, monitoring industrial digital systems, or participating in larger organizational risk-mitigation strategies.

Regarding course outcomes, I have already demonstrated proficiency in software engineering/design, algorithms and data structures, and the use of databases through my enhanced artifacts. I have also addressed testing, documentation, and security considerations. The remaining work involves final polish on the database category enhancement, updating the ePortfolio structure, and final validation of all artifacts prior to submission.

### **Part Two: Artifact Update**

| <b>Categories</b>             | <b>SDE</b>  | <b>Algorithm</b> | <b>Database</b> |
|-------------------------------|-------------|------------------|-----------------|
| <b>Name</b>                   | CS 360      | CS 300           | CS 360          |
| <b>Status of IE</b>           | Complete    | Complete         | In Progress     |
| <b>Submission status</b>      | Complete    | Complete         | In Progress     |
| <b>Status of FE</b>           | Complete    | Complete         | In Progress     |
| <b>Submitted to portfolio</b> | Complete    | Complete         | In Progress     |
| <b>Finalized Portfolio</b>    | In progress | In progress      | In Progress     |

### **Optional Feedback**

If possible, I would appreciate feedback on the clarity and completeness of the database enhancement explanation once I finalize it. Specifically, I want to ensure I adequately explain how the updated SQLite interactions strengthen security, improve reliability, and support the CS-499 outcome related to databases. I can provide links to the artifact after the last revisions if review is needed.